

C 243HS-443B-1: Healthcare Entrepreneurship, Fall 2024
The Heller School of Management and Health Policy
Brandeis University

Authors: Jimmy Belotte, Mariko Kita, Adnan Alseidi, Abhijai Singh



The Birth of a Revolution in Obstetrical Care MUMS

Maternal Unmanned Monitoring System



The Patient Perspective



The Provider
Perspective

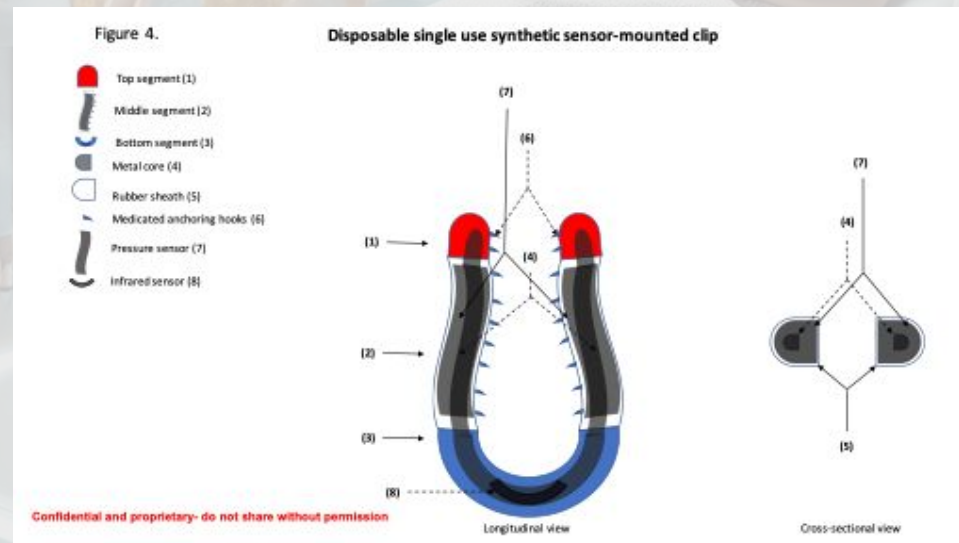
The Data and Safety Perspective



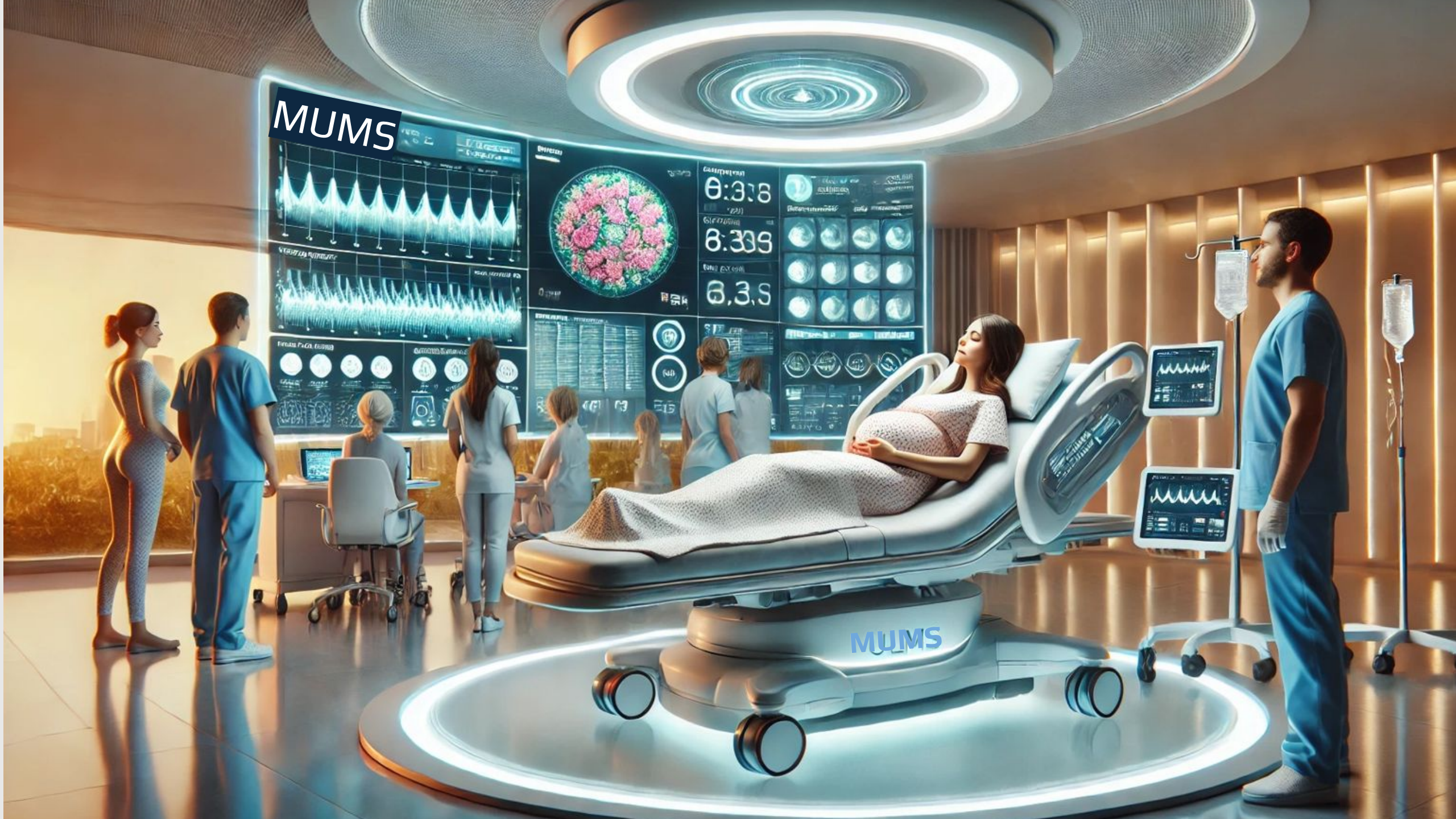


The Solution

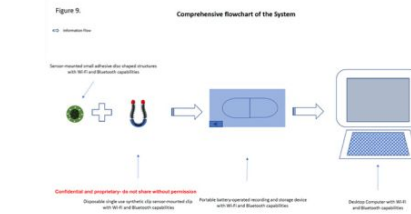
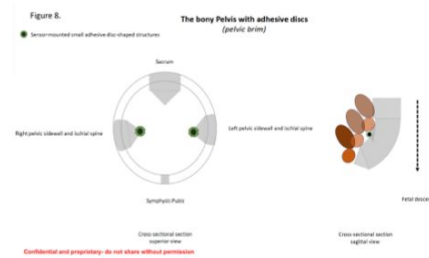
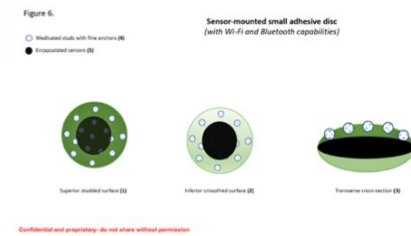
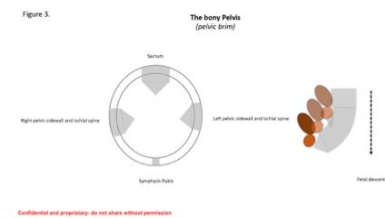
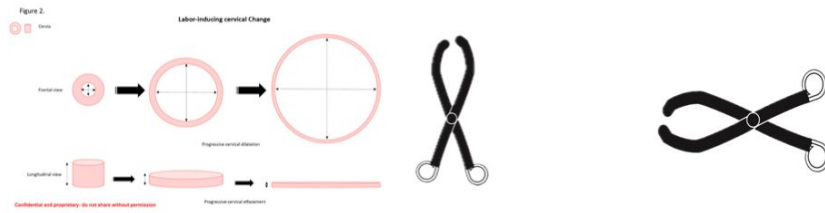
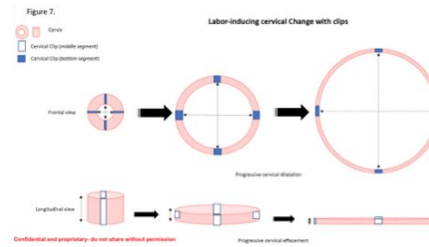
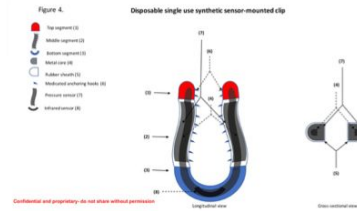
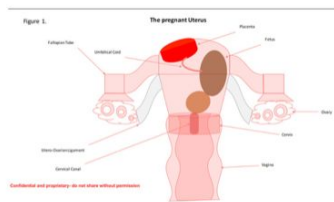
We are proposing the development and distribution of a sub-centimeter intravaginal device to continuously monitor the progress of obstetrical active labor which will capture, store, transfer, analyze and display key metrics related to the process of labor. Cervical changes, fetal presentation, station, position, uterine contraction dynamics, maternal and fetal vital signs as well as oxygen saturation can be measured.



MUMS



How it works



Social Impact-Value Proposition

Patient and Family

Potentially Safer
Less discomfort
Less invasive

Healthcare Providers

Assist HCPs so they can focus on
medical management
Increase confidence and well-being
Decrease risk and litigations
Increase work-life balance

Workforce

Partially address labor
shortage
Better predictable care
delivery and outcome

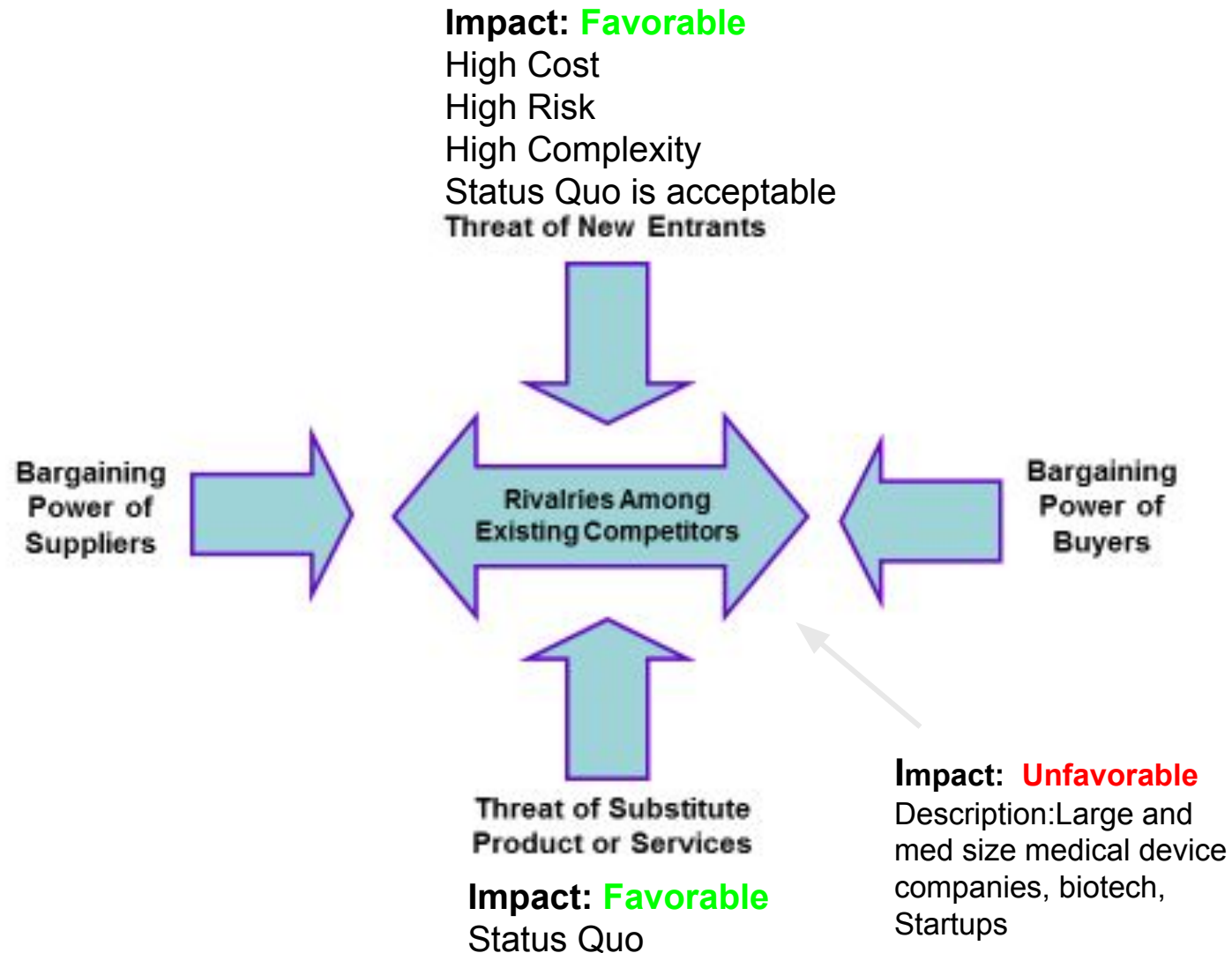
Investors

Favorable ROI, ESG
Contribute to improve
society

Our Ecosystem: Five forces

Impact: Unfavorable

Description:
Engineers
Scientists
Physicians
Vendors
Highly Specialized



Impact: Unfavorable

Description: Payers,
Hospital Systems,
Ambulatory centers,
Birthing units, OB.GYN

Competitors and Differentiation

Competitors:

- No competing devices at this time
- Established companies (GE, Hologic).
- Small biotech and startups.

Differentiation:

- Smart device
- Monitor cervical changes
- Risk stratification
- Intelligent Solution

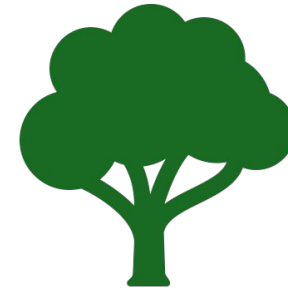
Ecosystem Overview



Partners:

NIH, State Programs, Academic Institutions,
Medical Societies.

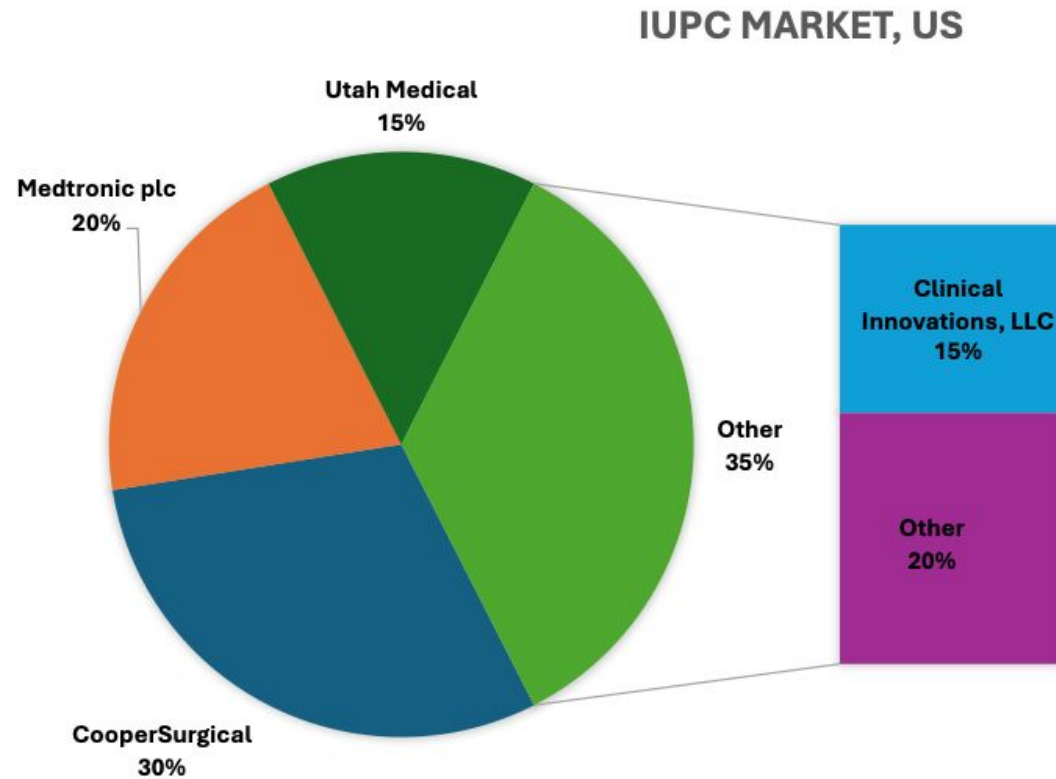
Philanthropic Organizations (e.g., March of
Dimes).



Ecosystem Complement:

Enhances existing practices such as natural
and home births.

Marketing: IUPC as a Proxy



- According to public reports by Verified Market Research
 - The IUPC market was valued at approximately **USD 170 million in 2023** and is projected to reach **USD 281 million by 2030**, growing at a **compound annual growth rate (CAGR) of 7.5%** during the forecast period from 2024 to 2030.
 - Another source, ReportPrime, indicates that the IUPC market is expected to grow from **USD 1.30 billion in 2022** to **USD 1.80 billion by 2030**, at a CAGR of 4.40% during the forecast period.

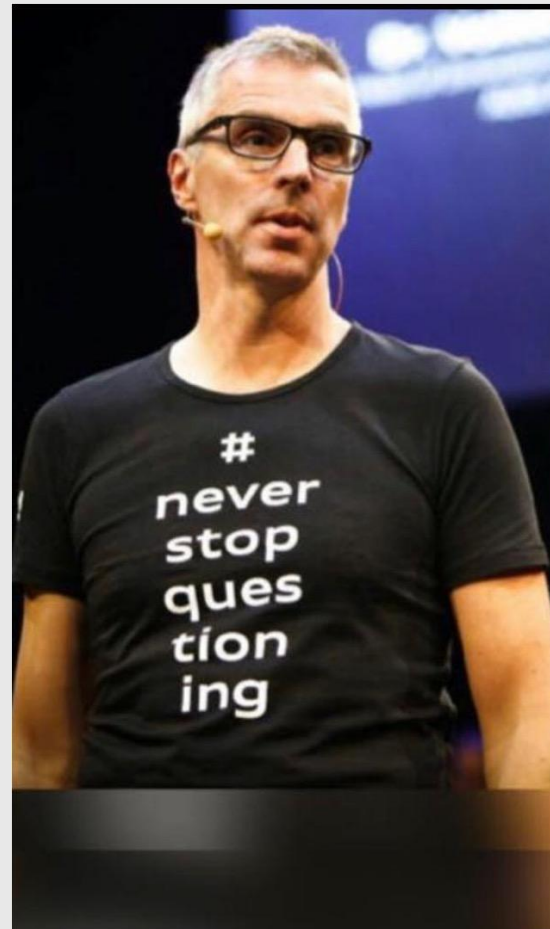
Meet the Team

Jimmy Belotte MD, PhD; FACOG



Founder/CEO
Board-certified OBGYN
NIH Scholar
Medical Director
GSK, Oncology Clin. Dev.
Formerly Associate
Professor
Albert Einstein College of
Medicine
Wayne State Univ. School of
Medicine

Volker Kease, PhD



CTO | Board Member | Strategic Advisor
| Investor | Former VW and Audi Head of
Innovation Management | Disruptive
mobility innovation

- Head of Project Volkswagen XL1 und XLSport | VOLKSWAGEN AG, Wolfsburg, D | 06-2013 to 10-2015
- Head of Quality Planning HV-Batteries and Chassis | VOLKSWAGEN AG, Braunschweig, D | 04-2011 to 05-2013
- Head of Materials Lab VW Group worldwide | VOLKSWAGEN AG, Wolfsburg | 07-2009 to 03-2011
- Personal Assistant to Chairman of the Board of Management of Volkswagen AG with responsibility for Research and Development | Volkswagen AG, Wolfsburg | 03-2007 to 06-2009
- Volkswagen Brand Manager for Innovation Management and Light Weight Components at Volkswagen Technical Development | Volkswagen AG, Wolfsburg, D | 2005 to 02-2007
- Head of several Light Projects | Volkswagen Group Research, Wolfsburg, D | 2002 to 02-2004
- Doctor of Engineering (Dr.-Ing.) in Materials Science, Diploma in Mechanical Engineering | University of Hanover, D | 1991 to 1997, to 2002
- Co-founder kw2-implants (resorbable metal surgery screws) | Hanover, D | 1996 to 2005
- Apprenticeship Deutsche Bank AG, Goslar, D | 1988 to 1990

Meet the Team

Ludwick Alexis, MBA



Director, Walmart International Finance
Walmart

Master of Business Administration June 2021

University of Chicago Booth School of Business, Chicago, IL

Master of Accounting May 2007

University of Florida, Gainesville, FL
Graduate Minor in Decision & Information Sciences

Bachelor of Science in Accounting May 2007

University of Florida, Gainesville, FL

CERTIFICATIONS

Certified Public Accountant – Licensed in Florida

PROFESSIONAL AFFILIATIONS

AICPA

LANGUAGES

Fluent in English and French

Edward Belotte, J.D.



Private Credit Associate at Katten Muchin Rosenman LLP.

2024-present

Capital Markets Associate

Dentons

Sep 2021 - Apr 2024

St. John's University School of Law
2018-2021

Legal Intern

Oppenheimer & Co. Inc.

Jan 2020 - May 2021

Judicial Intern

New York State Supreme Court

Jun 2020 - Sep 2020

Scaling and Growth Plan

- Secure an engineer who will build the device prototype
- Engage FDA early for Minimum Viable Product
- Create 20 prototypes
 - Outsource manufacturing of components
 - Assemble in Massachusetts
- Engage the early adopter user group of 10 doctors for feedback
- Publish results
- FDA: IDE/PMA

A futuristic hospital room with a pregnant woman lying in a bed. The bed is on a circular platform with the letters 'ULMS' on it. In the background, a large curved wall displays various medical data, including waveforms and graphs, with the letters 'ULMS' at the top. Several medical staff members in blue scrubs are standing around the room, and a man in blue scrubs is standing next to the bed. The room has a modern, clean design with large windows and a circular ceiling light fixture.

THANK YOU

Questions?

Appendices

510K Process for our device

Computerized Labor Monitoring Systems - Class II Special Controls Guidance for Industry and FDA Staff

Document issued on: April 24, 2007

For questions regarding this document, please contact Glenn Bell at 301-796-6531 or by email at glenn.bell@fda.hhs.gov (mailto:glenn.bell@fda.hhs.gov).



U.S. Department of Health and Human
Services
Food and Drug Administration
Center for Devices and Radiological
Health

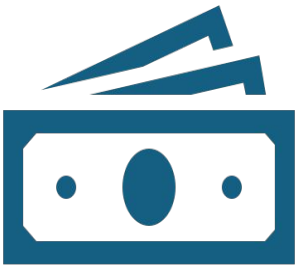
Obstetrics and Gynecological Devices Branch
Division of Reproductive, Abdominal, and
Radiological Devices
Office of Device Evaluation



Potential high level financial overview



“According a 2023 Forbes article, an average cost of \$18,865 (\$14,768 for vaginal and \$26,280 for cesarean) which included pregnancy, delivery and postpartum care.”
Annual birth rate in the US ~3.4 million



Projected Revenue Streams:

Year	1	2	3	4	5
#Unit sold	20.00	100.00	1000.00	10000.00	50000.00
Sales	\$10,000	\$50,000	\$500,000	\$5,000,000	\$25,000,000
COGS	\$ 4,000.00	\$ 20,000.00	\$ 200,000.00	\$ 2,000,000.00	\$ 10,000,000.00
GM	\$ 6,000.00	\$ 30,000.00	\$ 300,000.00	\$ 3,000,000.00	\$ 15,000,000.00
SG&A	\$ 3,500.00	\$ 17,500.00	\$ 175,000.00	\$ 1,750,000.00	\$ 8,750,000.00
Inc before Tax (25%)	\$ 2,500.00	\$ 12,500.00	\$ 125,000.00	\$ 1,250,000.00	\$ 6,250,000.00
Tax Exp	\$ 625.00	\$ 3,125.00	\$ 31,250.00	\$ 312,500.00	\$ 1,562,500.00
NP	\$ 1,875.00	\$ 9,375.00	\$ 93,750.00	\$ 937,500.00	\$ 4,687,500.00

Cost Structure: 80% Variable; 20% Fixed

Sale Price (Unit)	\$ 500.00
COGS (Unit)	\$ 200.00
GM (Unit)	\$ 300.00
SG&A (Unit)	\$ 175.00

Benefit/Risk Profile

The American Congress of Obstetricians and Gynecologists (ACOG) projects there will be a shortage of up to 8,800 obstetricians and gynecologists (OB-GYNs) by 2020, and a shortfall of up to 22,000 by 2050.

Benefit	Risk	Comment
Comprehensive labor monitoring	Pain on insertion	Local pain control
Continuous labor monitoring	Bleeding/spotting on insertion	negligible
Autonomous labor monitoring	Infection	Infection control, sterile application
Accuracy/objectivity	Slippage/falling out	Anchoring, engineering
Minimize discomfort	Data capture and transmission	Engineering/networking
Decrease infection	Cost	
Drug delivery		
Translational research		
Address workforce shortage		

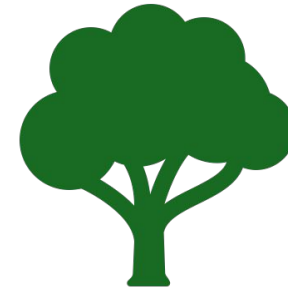
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